Yashvi Bulsara

yashvibulsara123@gmail.com

+491621586625

Yashvi Bulsara

EDUCATION

09/2024 – Present
Jena, Germany

Master of Science in Photonics, *Friedrich-Schiller-University*

08/2020 – 06/2024
Gandhinagar, India

Bachelor of Science (Hons.) in Physics,
Pandit Deendayal Energy University (NAAC A++)
Gold Medalist; CGPA : 9.90

EXPERIENCE

11/2025 – Present
Jena, Germany

Student Research Assistant,
Process Analysis of Pyrosil Films on Glass using Brewster Angle Reflectometry
Ongoing research at **INNOVENT e.V. – Technologieentwicklung, Jena**, in the *Magnetics and Optics Department (MOS)*, focusing on the characterization of Pyrosil coatings deposited via Catalytic Chemical Vapor Deposition (CCVD). The work involves performing optical measurements using Brewster Angle Reflectometry, followed by data processing and analysis to determine film thickness, refractive index, and surface properties, and to evaluate correlations between deposition parameters and optical performance.

04/2025 – Present
Jena, Germany

Master Student, *Hybrid; Transition Metal Dichalcogenides TMD/Nanowires NW; Field Effect Transistors FET*
I am currently interning at the **GUFOS group (IFK), FSU Jena,** where I work on the development of hybrid FETs by integrating 2D TMDs with 1D nanowires. The project combines the electrostatic gate control of TMDs with the high carrier mobility of nanowires and involves material synthesis, device fabrication, and characterization. This internship is also providing valuable hands-on experience in cleanroom processes and in building and testing nanoscale FET devices building and testing FETs.

RESEARCH & PROJECTS

07/2023 – 05/2024

Bachelor Thesis- Investigation of linear and non linear optical properties of ferrite based thin films.
I investigated Linear and Non- Linear Optical Properties of Ferrite-based device and films. I investigated monolayered and multi-layered thin films of Ferrite-based devices. The observed non-linearity could be considered as an useful application in optoelectronics in the future. I used Z-Scan, UV-Visible Spectroscopy and other techniques which were available at the institute.

01/2022 – 01/2023

Laser Induced Breakdown Spectroscopy for the analysis of trace metals in liquid.
For this project, Pandit Deendayal Energy University granted our team a **research grant of INR 75,000** for a year under the Student Research Programme; Office of Dean R&D. To explore the field of Enhanced LIBS we upgraded the normal LIBS setup which is generally used; by adding a Nebuliser. We applied the principle of Aerosol LIBS to the experimental setup because liquid samples were involved.

CONFERENCE PRESENTATIONS

03/2026 – 03/2026
Dresden, Germany

Optical Properties of a Transition-Metal Dichalcogenide ZnO Nanowire Field-Effect Transistor (Poster),
DPG Spring Meeting 2026 – Deutsche Physikalische Gesellschaft

- Presented a research poster on hybrid ZnO nanowire–WSe_2 field-effect transistors.
- Investigated photoluminescence and gate-dependent optical behavior in 1D–2D semiconductor heterostructures.
- Demonstrated device fabrication, electrical characterization, and optical measurements for nanoscale photonic devices.
- Discussed the potential of hybrid nanostructures for electrically tunable nanophotonic applications.

WORKSHOPS & SUMMER SCHOOL

09/2025 – 09/2025
Freiburg, Karlsruhe & Aachen, Germany

Fraunhofer Photonica Summer School
Participated in the *Fraunhofer Photonica 2025 Summer School*, hosted by **Fraunhofer IPM**, **IOSB**, and **ILT**. The program covered laser-based light sources, LIDAR, holography, nonlinear optics, optical metrology, infrared technologies, crystal growth, and quantum photonics, combining lectures with lab demonstrations and discussions with Fraunhofer researchers. Gained hands-on experience in optical inspection, spectroscopic techniques, and laser applications, and developed a deeper understanding of how photonics research is translated into industrial and scientific innovations.

TECHNICAL SKILLS

Programming Languages: Python, C

Softwares: Origin, QGIS, Zemax Optic Studio, OpenGrADS, Arduino IDE, MATLAB

Techniques: Z-Scan system, UV-Visible Spectroscopy, Raman Spectroscopy, Fourier Transform Infrared Spectroscopy (FTIR), Scanning Electron Microscopy (SEM) , X-Ray Diffraction (XRD)

Documentation: LaTeX, MS Office 365

Soft Skills: Leadership, Organizational Skills, Communication Skills, Time Management, Analytical Thinking and Problem Solving

POSITIONS OF RESPONSIBILITY

12/2022 – 12/2024 **Board of Studies; Student Member,** *Department of Physics, PDEU*

05/2021 – 12/2023 **Editorial Team,** *Physitien-Physics Newsletter, PDEU*

06/2022 – 06/2023 **President,** *Nucleus-The Physics Club of PDEU, PDEU*

08/2021 – 06/2022 **Operations Head,** *Nucleus-The Physics Club of PDEU, PDEU*

09/2022 – 09/2022 **Student Volunteer,**
International Conference on Condensed Matter and Device Physics (2023), PDEU

School Council, *Shree Vallabh Ashram MGM and VN Savani School*

LANGUAGES

English: C1; **IELTS**:7.5 | **Hindi** | **Gujarati** | **German:** A2 | **French:** A2